Deccan AI Campus, Hall A: electrical and liquid-cooling design-build, technical requirements

Client: Deccan Compute (fictional)
Site: AI campus near Hyderabad (fictional)
Reference: DC-HA-TS-001 Rev 0
Fictional tender authored for the Kirchhoff demo. IEC-based, illustrative, not for construction.

§1.1 Scope of works

The contractor shall design, supply, install and commission the electrical power distribution and the direct liquid cooling for Data Hall A, from the 33 kV main switchboard to the rack power inlets and the rack coolant manifolds.

§2.3 Programme

All major electrical and mechanical equipment shall be on site no later than 26 weeks from contract award. Equipment that cannot be delivered within this period shall not be specified.

§3.1 IT load

The hall shall accommodate 48 racks in 6 rows of 8, each rack rated 132 kW. For design purposes 80% of rack heat is rejected to liquid and 20% to air.

§3.2 Auxiliary load

Auxiliary load on the LV system, including CDU pumps, fan walls and controls, shall be taken as 0.30 MW in total. CDU pumps are UPS-backed; other auxiliary loads are not.

§3.3 Electrical supply

Supply is at 33 kV, 50 Hz, from a main switchboard with two incomers and an automatic bus-coupler; the source fault level is 1,000 MVA with R/X 0.1. Distribution is at 415 V with a design power factor of 0.95.

§4.2 Redundancy

The critical power distribution shall be 2N, with each rack fed from two independent systems.

§4.3 Transformer loading

Transformer loading shall not exceed 80% of rating in normal operation and 100% of rating under any single failure.

§4.4 Short-circuit ratings

Switchgear and busways shall be rated for the maximum prospective short-circuit current at their point of installation.

§4.5 UPS

Each UPS system shall carry its worst single-failure load with a battery autonomy of not less than 5 minutes.

§4.6 Voltage drop

The voltage drop from the UPS output board to the furthest busway tap-off shall not exceed 2.0% at the worst-case design current.

§5.1 Containment tiers

Overhead services shall run on three tiers: low-level pipework at 2.6 m, busways at 3.2 m and mechanical mains at 3.8 m above finished floor.

§6.1 Coolant

Coolant supply temperature shall be 30 °C with a design temperature rise of 10 K across each rack.

§6.2 CDU redundancy and pressure

CDUs shall be provided N+1 per hydraulic zone. The index-circuit pressure drop shall be within the CDU available external head.

§6.3.4 Liquid above electrical

No liquid-carrying pipework shall be routed above electrical busways or feeders. Pipework shall keep a clearance of at least 300 mm from structural columns.

§6.4 Velocity

Coolant velocity in any pipe shall not exceed 2.5 m/s.

§6.5 Continuous cooling

CDU pumps shall be fed from two independent UPS-backed supplies to maintain continuous cooling.

App. B Appendix B: acceptable alternatives

Block or distributed redundancy acceptable, subject to the loading limits of §4.3.